
Scalable Parallel-in-Time Integration

Abstract
stuff

1. INTRODUCTION

- The Equation of Motion
- Acoustics in Expanding Volumes

2. METHODS

A. THE PARAREAL ALGORITHM

- "Parallel-in-time integration" $\approx$ "Solving IVP in parallel"
- Several algorithms
 - Parareal
 - Multigrid Reduction in Time (MGRIT)
 - Parallel Full Approximation Scheme in Space and Time (PFASST)
 - and more
- 4 core steps
 - Prepare the subproblems
 - Solve each subproblem
 - Correct
 - Iterate

i. Subproblem Preparation

Given an initial value problem P

$$\mathcal{L}(t, u, \partial_t u, \partial_t^2 u) = f(t) \quad u(0) = u_0, \partial_t u(0) = v_0 \quad D = [T, T + \Delta T] \quad (1)$$

Choose number of subproblems N (suggest number of compute cores)

Partition time domain for $p = 0, 1, \dots, N - 1$

$$D = [T, T + \Delta T] \rightarrow D_p = [T + p/N\Delta T, T + (p + 1)/N\Delta T] = [T_p, T_{(p + 1)}] \quad (2)$$

Choose coarse propagator $\mathcal{C}_0$ to get initial solution values $\{u_p^0\}_p, \{v_p^0\}_p$

$$P^0 \equiv \mathcal{L}(t, u, \partial_t u, \partial_t^2 u) = f(t) \quad \underbrace{u(0) = u_0, \partial_t u(0) = v_0}_{\text{Subproblem initial values}} \quad \underbrace{D = [T, T + \Delta T]}_{\text{Subproblem domain}} \quad (3)$$

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- ii. Parallel Propagation
 - iii. Corrections
 - iv. Iterate
 - B. GPU COMPUTING
 - C. DISTRIBUTED COMPUTING
 - 3. RESULTS
 - 4. DISCUSSION
 - 5. CONCLUSION
 - A. FUTURE WORK
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